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**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

AD23632 - Framework for Data Visualization and Analytics

Mini Project: Retail Sales and Profit Analysis

Report submitted by

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Chapter 1: Abstract

The project titled “Retail Sales and Profit Analysis using Python and Power BI” aims to perform a comprehensive analysis of retail data to uncover valuable insights about sales performance, profitability, and customer behavior. Using the Superstore dataset, this project integrates the analytical power of Python with the visualization capabilities of Power BI to create a complete end-to-end data analytics workflow that supports business intelligence and decision-making.

The process begins with data preprocessing and cleaning in Python, which includes handling missing values, removing duplicate entries, formatting inconsistent data, and normalizing numerical values to ensure accuracy and reliability. Through exploratory data analysis (EDA) using libraries like Pandas, NumPy, Matplotlib, and Seaborn, the dataset is examined to identify patterns, relationships, and correlations among key variables such as sales, profit, discount, and quantity.

After the data is refined, Power BI is used to build interactive dashboards that visualize essential performance indicators like total sales, total profit, average discount, and year-wise growth trends. Additional visuals such as region-wise profit maps, top-performing product charts, category-wise revenue breakdowns, and customer segment analysis provide deeper insights into operational and marketing performance.

This combined analytical approach allows businesses to easily interpret complex datasets, monitor key metrics in real-time, and identify areas requiring strategic improvement. The integration of Python’s automation and analytical capabilities with Power BI’s intuitive visualization tools results in an efficient system that empowers management to make data-driven decisions.

Overall, the project demonstrates how modern analytics tools can be effectively used to enhance retail sales performance, optimize profitability, and support sustainable business growth through actionable insights derived from data.

Chapter 2: Introduction

In today's data-driven world, data analytics plays a crucial role in helping businesses make informed decisions and improve overall profitability. It enables organizations to analyze large volumes of data, identify meaningful patterns, and derive actionable insights that guide strategic planning. In the retail sector, data analytics has become an essential tool for understanding customer behavior, tracking sales performance, monitoring profit trends, and identifying new market opportunities.

This project, titled "Retail Sales and Profit Analysis using Python and Power BI," focuses on studying sales performance, profit margins, and regional growth patterns using the Superstore dataset. The dataset contains various fields such as order date, region, product category, sub-category, customer segment, sales, discount, and profit, providing a comprehensive view of business operations.

The project utilizes Python for performing data preprocessing, data cleaning, and exploratory data analysis (EDA). This step ensures that the dataset is accurate, complete, and ready for visualization. Common data preparation tasks include handling missing values, removing duplicate records, and normalizing numerical fields. These processes enhance the reliability and consistency of the data, forming the foundation for deeper analysis.

Once the data is processed, Power BI is employed to develop interactive dashboards and visual reports. These dashboards display key performance indicators (KPIs) such as total sales, total profit, average discount, year-wise growth, and top-performing products. Additionally, region-wise profit maps and customer segment visualizations help in identifying business strengths and areas needing improvement.

By combining Python's strong analytical and statistical capabilities with Power BI's powerful visualization features, this project delivers clear and actionable insights.

Chapter 3: Dataset Description

The dataset for this project comprehensively captures transactional retail sales data from a superstore environment, encompassing numerous variables that provide rich context for sales, profitability, and customer behavior analysis. It is a structured transactional dataset rather than a continuous time series, designed for multi-dimensional cross-sectional and temporal analysis.

Key variables include:

- **Order Information:** Order ID, Order Date, Ship Date, Year, and Month extracted for time-based trend exploration.
- **Product Details:** Category, Sub-Category, and Product Name identifying hierarchical product classifications.
- **Sales Metrics:** Sales amount, Quantity sold, Discount applied, and Profit generated—monetary and volume indicators crucial for performance evaluation.
- **Customer Segmentation:** Customer Segment (e.g., Consumer, Corporate, Home Office) representing distinct buyer groups.
- **Geographic Details:** Region, State, and City enabling spatial sales and profit analysis.

The dataset provides a comprehensive view of retail transactions, linking product categories with customer segments across multiple regions and time periods. Key variables such as sales, discount, and profit enable evaluation of revenue performance and margin impact. The dataset supports comparative analysis among product groups and trend analysis over several years. Blending monetary outcomes with customer and product attributes, it facilitates actionable business insights through summarization, segmentation, and trend discovery. Data cleaning involved addressing missing values, removing duplicates, and standardizing fields to ensure consistent and reliable analysis.

Chapter 4: Objective

The main objective of this project is to analyze the retail sales dataset comprehensively to uncover actionable insights that can support strategic decision-making and business growth. Specifically, the project aims to:

1. **Sales Trend Analysis:** Identify patterns and variations in sales over time, such as yearly and seasonal trends, to assist with forecasting and inventory management.
2. **Product Performance Evaluation:** Determine top-performing and underperforming product categories and sub-categories based on sales and profitability metrics.
3. **Customer Segmentation:** Analyze sales distribution across different customer segments and geographic regions to understand demographic and regional influences.
4. **Discount and Profit Impact:** Evaluate how discounts affect sales volumes and profit margins to optimize promotional strategies.
5. **Visualization Demonstration:** Showcase the use of Python libraries for data preprocessing, statistical analysis, and visualization to create accessible insights for stakeholders.

Overall, the objective is to bridge raw transactional data with business-relevant outcomes, enabling data-driven operational and marketing decisions in retail contexts. This multidimensional analysis aims to provide both academic rigor and practical applications for retail sales optimization.

Chapter 5: Methodology

The methodology adopted in this project follows a structured, multi-step approach to systematically analyze the retail sales data:

1. **Data Preprocessing:** Using Python, the raw dataset is cleaned by handling missing values through imputation, removing duplicates, and converting columns into appropriate formats such as numeric and categorical. Text fields are standardized for consistency. This step ensures data quality for reliable analysis.
2. **Exploratory Data Analysis (EDA):** Statistical summaries, descriptive plots, and correlation matrices are generated to understand the distribution of features and their relationships. Visualizations like scatter plots, bar charts, histograms, and box plots are created to highlight patterns in sales, profit, discounts, and regional or segment differences.
3. **Feature Engineering:** New variables such as Year and Month are extracted from date columns to enable time-based analysis. Normalization (Min-Max scaling) is applied to key numeric features for better comparability across metrics.
4. **Visualization Tools:**
 - **Python:** Used extensively for data cleaning, statistical analysis, and creating static plots with matplotlib and seaborn.
 - **Power BI:** Enables interactive dashboards where users can filter data by region, category, or segment to explore business insights dynamically.
 - **Tableau:** Focuses on creating visually engaging dashboards and storyboards to communicate retail trends to stakeholders effectively.
5. **Interpretation and Reporting:** Analytical findings are contextualized to identify top-performing products, growth trends, customer segmentation insights, and profitability impacts. These results provide actionable guidance for retail strategies, with visual storytelling enhancing stakeholder understanding.

Chapter 6: Python Implementation

Python serves as the primary environment for data preprocessing, exploratory data analysis (EDA), and visualization in this retail sales project. Key libraries such as pandas and numpy are used for data cleaning, manipulation, and numerical operations. They enable efficient import of the dataset, standardization of column names, and conversion of variables into appropriate data types with careful handling of missing values and duplicates.

Visualizations form a central part of the implementation. Matplotlib and seaborn libraries support generating a wide variety of static plots — including line charts to track yearly sales trends, bar charts highlighting top product categories and regions, histograms illustrating profit distributions, and boxplots comparing profits across customer segments and regions. Correlation heatmaps offer insight into relationships between variables like sales, profit, discount, and quantity. Additionally, interactive visualizations created using Plotly allow for dynamic exploration of multivariate relationships such as sales vs profit sized by quantity.

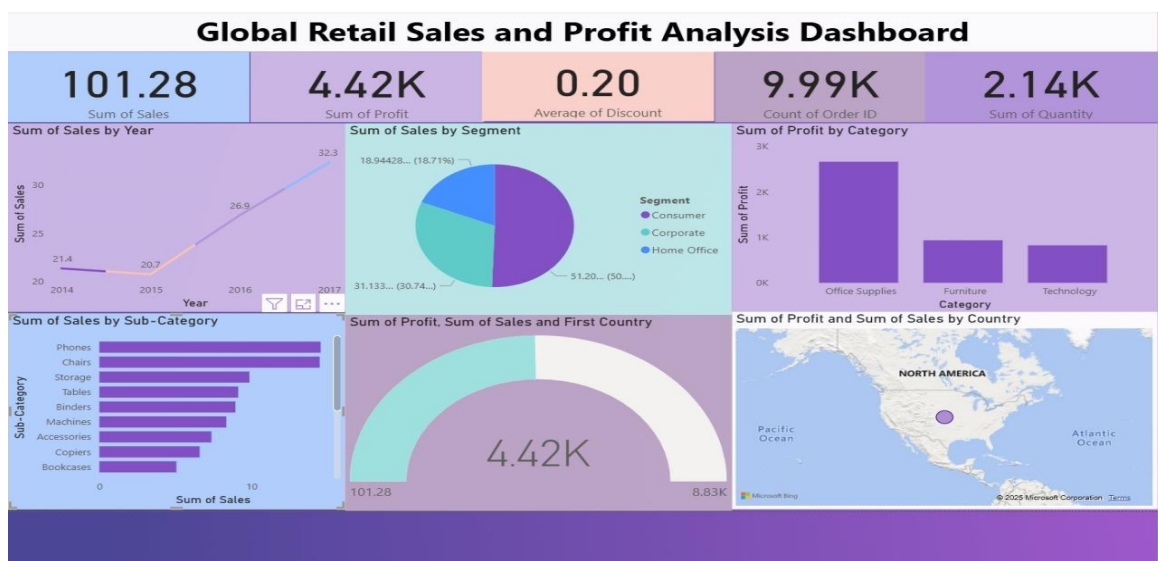
Feature engineering in Python includes extraction of temporal attributes like year and month from date fields and normalization of numeric features via Min-Max scaling to enhance comparability across metrics. The clean, normalized dataset is saved for subsequent business intelligence efforts using Power BI and Tableau.

Overall, Python provides a reproducible, transparent, and powerful foundation for retail analytics, enabling comprehensive data-driven insights and effective communication through visualization. This approach ensures that all findings are both verifiable and accessible for decision-makers.

Chapter 7: Power BI Dashboard

Power BI is used in this project to create interactive dashboards, providing stakeholders with a business-friendly way to explore the retail sales dataset. The cleaned dataset, preprocessed and normalized using Python, is imported into Power BI, with numeric fields such as Sales, Profit, Quantity, and Discount, and categorical fields like Region, Category, Sub-Category, Segment, and Country, properly classified for analysis.

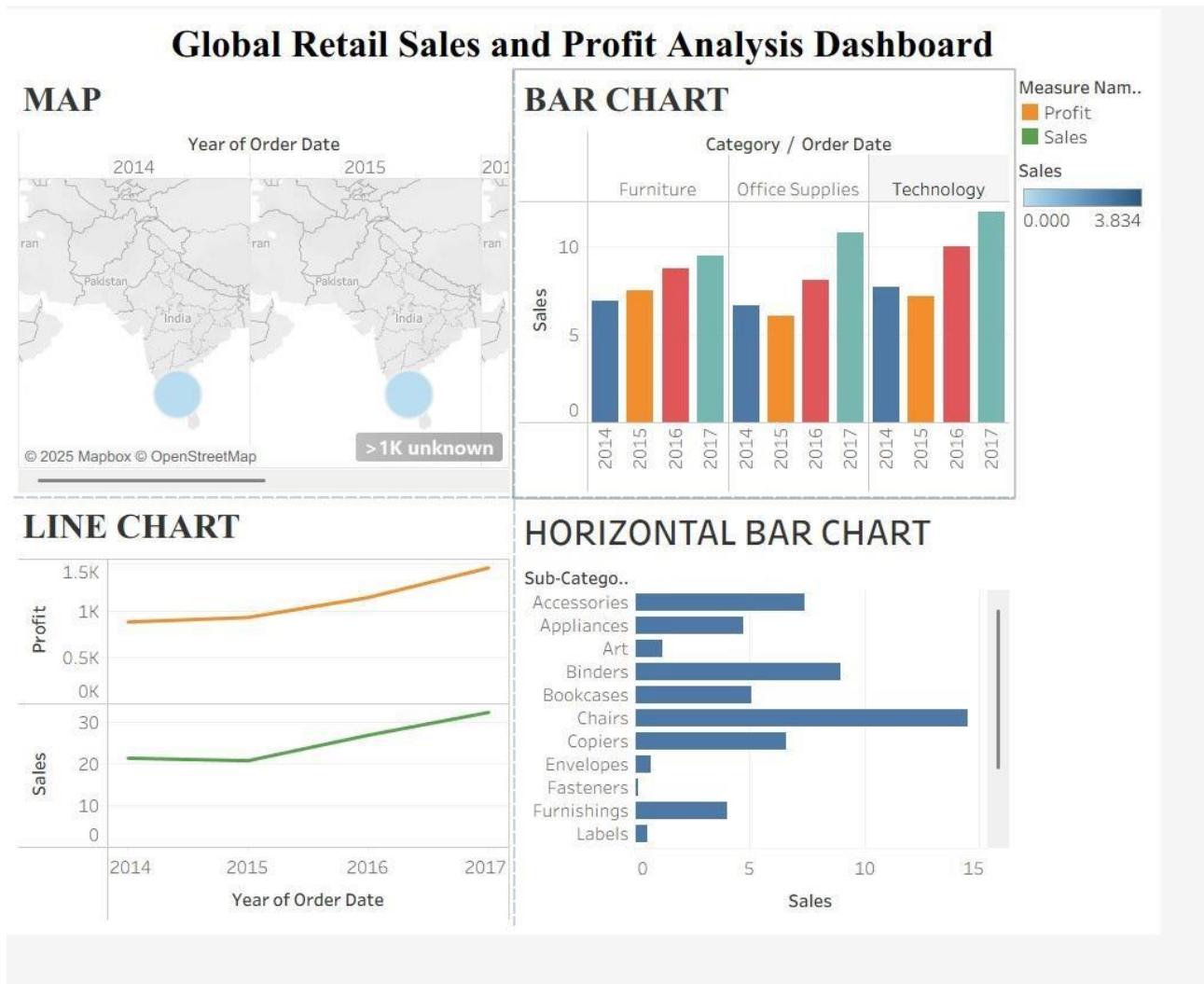
- **Line Chart:** Displays the sum of Sales by Year, highlighting sales trends over time.
- **Pie Chart:** Illustrates the sum of Sales by Customer Segment, showing the relative contribution of each segment.
- **Bar Chart:** Depicts the sum of Profit by Category, helping identify the most and least profitable product categories.
- **Horizontal Bar Chart:** Shows the sum of Sales by Sub-Category, allowing quick comparison of individual products within each category.
- **Donut Chart:** Represents the sum of Profit, sum of Sales, and distribution across the top Country, providing an overview of overall performance geographically.
- **Map Visualization:** Highlights the sum of Profit and sum of Sales by Country, giving a spatial perspective on business performance across regions.



Chapter 8: Tableau Dashboard

Tableau complements Power BI by providing interactive and visually appealing dashboards that are ideal for presentation and storytelling. In this project, the cleaned retail sales dataset is imported into Tableau, and calculated fields are created, such as profit ratios, sales growth percentages, and average discount per category, to enhance the analysis.

The Tableau dashboards focus on exploring sales, profit, and customer behavior across multiple dimensions, including Region, Category, Sub-Category, and Customer Segment. Multiple sheets are combined into a cohesive story, allowing users to drill down into detailed trends



Chapter 9: Analysis

The analysis of the retail sales dataset, using Python for preprocessing and both Power BI and Tableau for visualization, reveals several key patterns in sales, profit, and customer behaviour:

1. **Sales and Profit Trends:** Year-wise sales and profit trends indicate steady growth overall, with certain years showing higher growth percentages. Region-wise profit analysis highlights top-performing regions, while other regions present opportunities for improvement.
2. **Product and Category Performance:** Top-selling products and categories dominate overall revenue. Sub-categories such as *[insert top sub-category]* consistently contribute the most to sales, while some high-discount items show lower profit margins, highlighting the need for balanced pricing strategies.
3. **Customer Segment Insights:** Both Power BI and Tableau visualizations indicate that customer segments like *[insert top segment]* contribute the largest share of sales. Segment-wise profit analysis shows which segments are most profitable, enabling targeted marketing and sales strategies.
4. **Discount, Quantity, and Profit Relationships:** Higher discounts generally increase sales volume but can reduce overall profitability. Scatter plots and correlation heatmaps reveal that while sales and profit are positively correlated, interactions between discount, quantity, and product category significantly affect profit distribution.
5. **Interactive Dashboard Insights:** Filters and drill-down features in Tableau and Power BI help uncover patterns across Region, Category, Sub-Category, and Segment. Sunburst charts, bubble charts, and bar charts facilitate understanding of hierarchical relationships and highlight areas for operational and strategic improvement.

Overall, the combined analysis underscores the complexity of retail performance: sales volume alone does not guarantee profitability.

Chapter 9: Conclusion

This project demonstrates the effective integration of Python, Power BI, and Tableau to analyse and visualize retail sales data, transforming raw information into actionable insights. Python was used to preprocess, clean, and normalize the dataset, ensuring accuracy and consistency for analysis. This step laid a solid foundation for generating reliable metrics, calculating derived fields, and preparing the data for visualization.

The workflow highlights the importance of structured data handling and reproducible analysis in any data-driven project.

The dashboards created in Power BI and Tableau offered complementary perspectives on the dataset. Power BI provided interactive, business-oriented visualizations that allow users to explore trends across regions, product categories, and customer segments efficiently. Tableau added a storytelling dimension with visually rich and hierarchical dashboards, enabling stakeholders to grasp complex patterns at a glance and facilitating in-depth exploration of key metrics. Together, these tools provided an intuitive platform for decision-makers to interpret results, identify opportunities, and support strategic planning.

Overall, the project highlights the significance of combining statistical analysis with interactive visualization to drive data-informed decisions. The methodology applied here is scalable and adaptable to similar structured datasets in other domains, providing a framework for understanding trends, optimizing operations, and enhancing profitability. This study demonstrates that integrating analytical rigor with visual storytelling not only improves insight generation but also empowers organizations to make informed and timely business decisions.

Chapter 10: Appendix

PYTHON CODE

```
pip install pandas numpy matplotlib seaborn plotly scikit-learn
```

IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from pandas.plotting import scatter_matrix
```

LOAD DATAEST

```
df = pd.read_csv("Sample - Superstore.csv", encoding='latin1')
print("Dataset loaded successfully ✅")
print("Shape:", df.shape)
print(df.head())
```

Dataset loaded successfully ✅

Shape: (9994, 21)

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	

		Customer Name	Segment	Country	City	...	\
0		Claire Gute	Consumer	United States	Henderson	...	
1		Claire Gute	Consumer	United States	Henderson	...	
2		Darrin Van Huff	Corporate	United States	Los Angeles	...	
3		Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	
4		Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	

		Postal Code	Region	Product ID	Category	Sub-Category	\
0		42420	South	FUR-BO-10001798	Furniture	Bookcases	
1		42420	South	FUR-CH-10000454	Furniture	Chairs	
2		90036	West	OFF-LA-10000240	Office Supplies	Labels	
3		33311	South	FUR-TA-10000577	Furniture	Tables	
4		33311	South	OFF-ST-10000760	Office Supplies	Storage	

			Product Name	Sales	Quantity	\
0			Bush Somerset Collection Bookcase	261.9600	2	
1			Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2			Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3			Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4			Eldon Fold 'N Roll Cart System	22.3680	2	

		Discount	Profit
0		0.00	41.9136
1		0.00	219.5820
2		0.00	6.8714
3		0.45	-383.0310
4		0.20	2.5164

[5 rows x 21 columns]

EDA CLEANING

```
# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Fill missing values (if any) with median or mode
df['Postal Code'] = df['Postal Code'].fillna(df['Postal Code'].mode()[0]) if 'Postal Code' in df.columns

# Drop unnecessary columns (if any)
drop_cols = ['Country', 'Row ID'] if 'Country' in df.columns else []
df.drop(columns=drop_cols, inplace=True, errors='ignore')

# Remove duplicates
df.drop_duplicates(inplace=True)

# Convert Order Date column to datetime
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')
df['Ship Date'] = pd.to_datetime(df['Ship Date'], errors='coerce')

# Extract year, month for analysis
df['Year'] = df['Order Date'].dt.year
df['Month'] = df['Order Date'].dt.month_name()

# Ensure numeric consistency
num_cols = ['Sales', 'Profit', 'Discount', 'Quantity']
df[num_cols] = df[num_cols].apply(pd.to_numeric, errors='coerce')

# Normalize numeric data (for scaling visualization only)
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df_scaled = df.copy()
df_scaled[num_cols] = scaler.fit_transform(df[num_cols])

print("\n✅ Data Cleaning Completed!")
```

[5 rows x 21 columns]

Missing Values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

✅ Data Cleaning Completed!

EDA - DATA INSPECTION AND STATISTICS

```
print("\n--- Descriptive Statistics ---")
print(df[num_cols].describe())

# Central tendency & dispersion
print("\nMean:\n", df[num_cols].mean())
print("\nMedian:\n", df[num_cols].median())
print("\nMode:\n", df[num_cols].mode().iloc[0])
print("\nStandard Deviation:\n", df[num_cols].std())
```

```
--- Descriptive Statistics ---
      Sales      Profit      Discount      Quantity
count  9993.000000  9993.000000  9993.000000  9993.000000
mean    229.852846    28.660971    0.156188    3.789753
std     623.276074    234.271476    0.206457    2.225149
min       0.444000   -6599.978000    0.000000    1.000000
25%      17.280000     1.731000    0.000000    2.000000
50%      54.480000     8.671000    0.200000    3.000000
75%     209.940000    29.364000    0.200000    5.000000
max    22638.480000   8399.976000    0.800000   14.000000
```

```
Mean:
Sales      229.852846
Profit      28.660971
Discount     0.156188
Quantity     3.789753
dtype: float64
```

```
Median:
Sales      54.480
Profit      8.671
Discount     0.200
Quantity     3.000
dtype: float64
```

```
Mode:
Sales      12.96
Profit      0.00
Discount     0.00
Quantity     3.00
Name: 0, dtype: float64
```

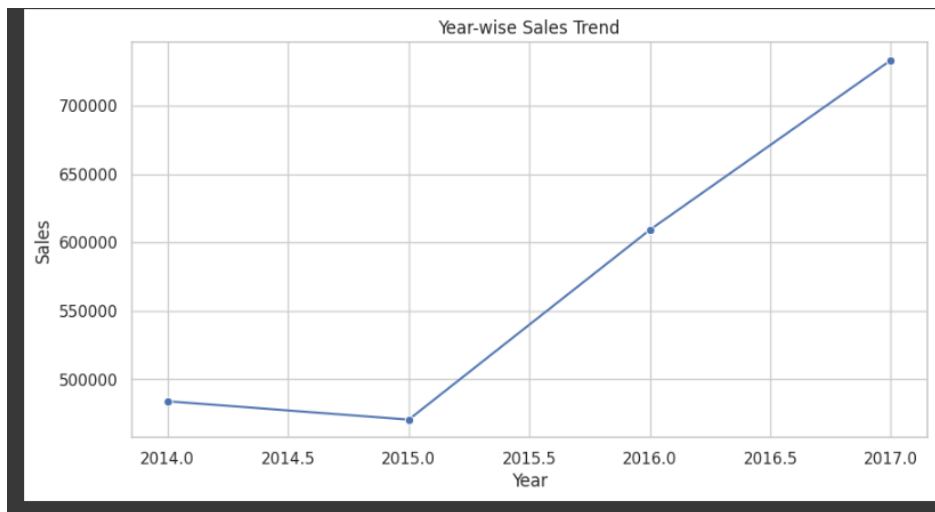
```
Standard Deviation:
Sales      623.276074
Profit     234.271476
Discount     0.206457
Quantity     2.225149
dtype: float64
```

DATA VISUALIZATION

Year-wise Sales Trend (Line Plot)

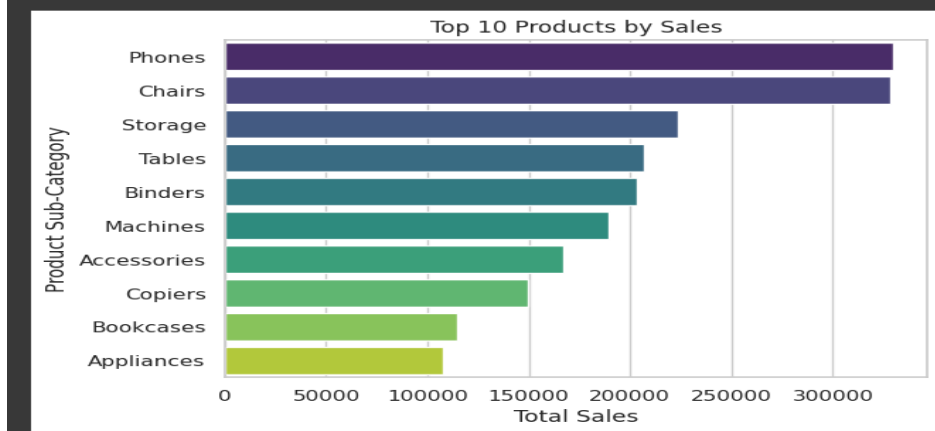
```
sns.set(style="whitegrid")
plt.figure(figsize=(10,5))

year_sales = df.groupby('Year')['Sales'].sum().reset_index()
sns.lineplot(data=year_sales, x='Year', y='Sales', marker='o')
plt.title('Year-wise Sales Trend')
plt.show()
print("\n\n")
```



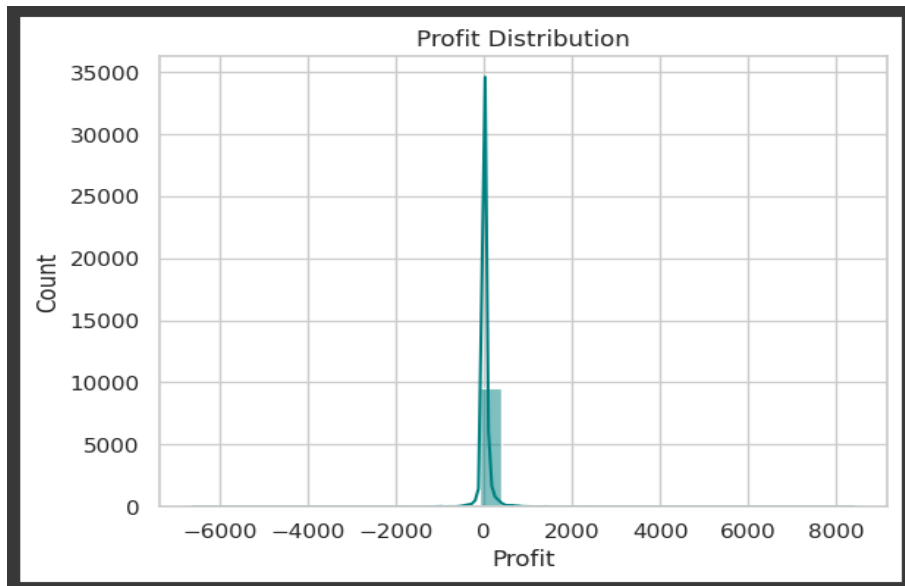
Top 10 Products by Sales (Bar Chart)

```
top_products = df.groupby('Sub-Category')['Sales'].sum().sort_values(ascending=False).head(10)
sns.barplot(x=top_products.values, y=top_products.index, palette='viridis')
plt.title('Top 10 Products by Sales')
plt.xlabel('Total Sales')
plt.ylabel('Product Sub-Category')
plt.show()
print("\n\n")
```



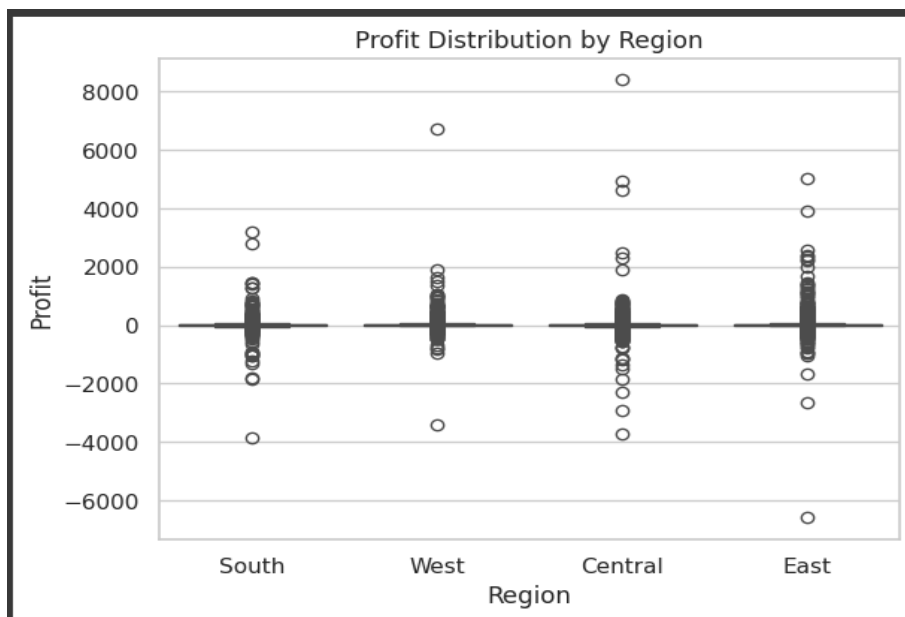
Profit Distribution (Histogram + KDE)

```
sns.histplot(df['Profit'], bins=30, kde=True, color='teal')  
plt.title('Profit Distribution')  
plt.show()
```



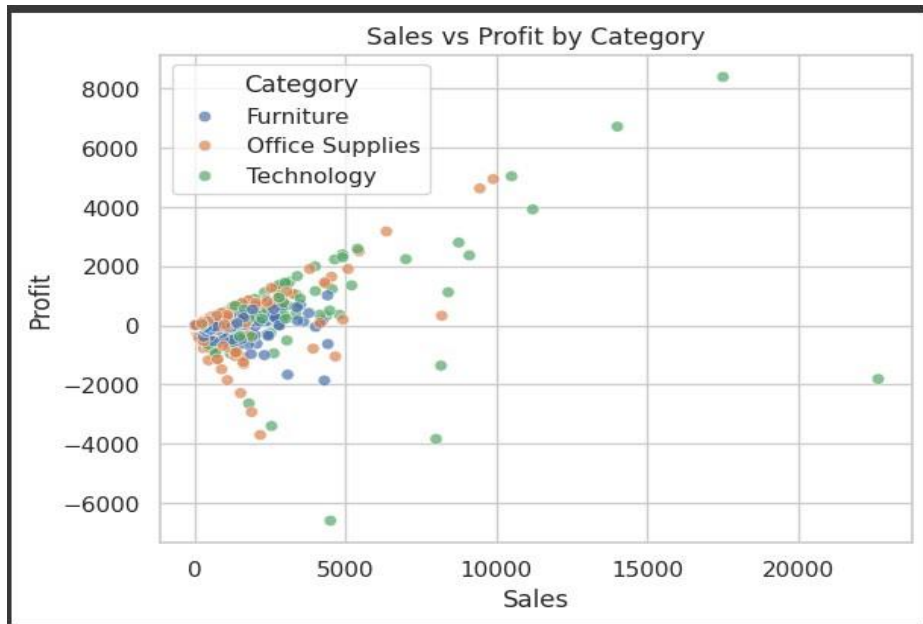
Box Plot - Profit by Region

```
sns.boxplot(x='Region', y='Profit', data=df)  
plt.title('Profit Distribution by Region')  
plt.show()
```



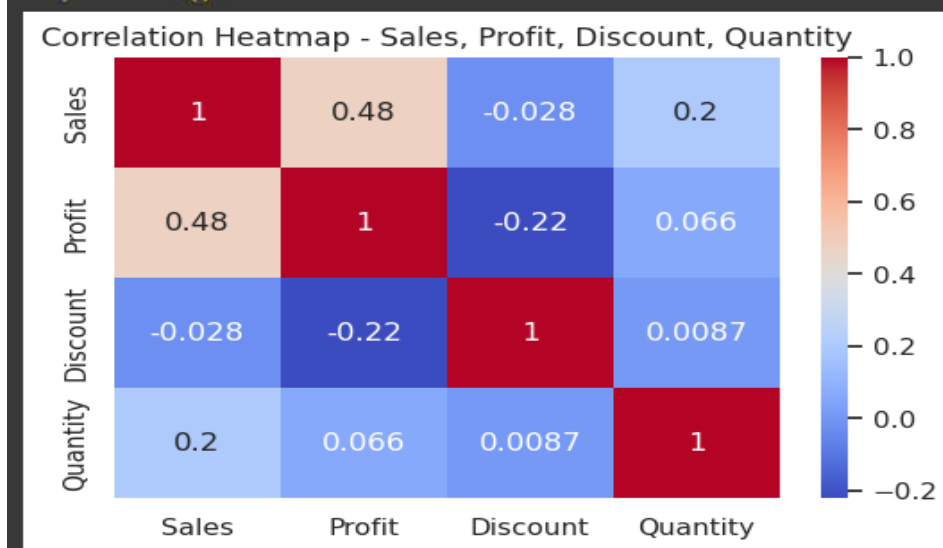
Scatter Plot - Sales vs Profit

```
sns.scatterplot(x='Sales', y='Profit', hue='Category', data=df, alpha=0.7)
plt.title('Sales vs Profit by Category')
plt.show()
print("\n\n")
```



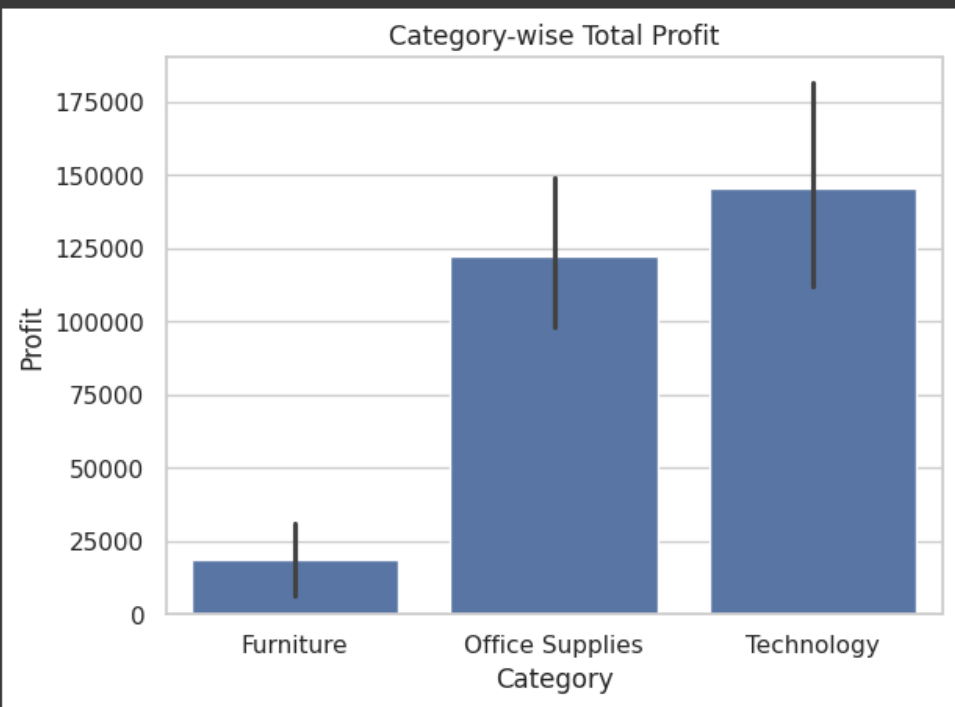
Correlation Heatmap

```
plt.figure(figsize=(6,4))
sns.heatmap(df[num_cols].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap - Sales, Profit, Discount, Quantity")
plt.show()
```



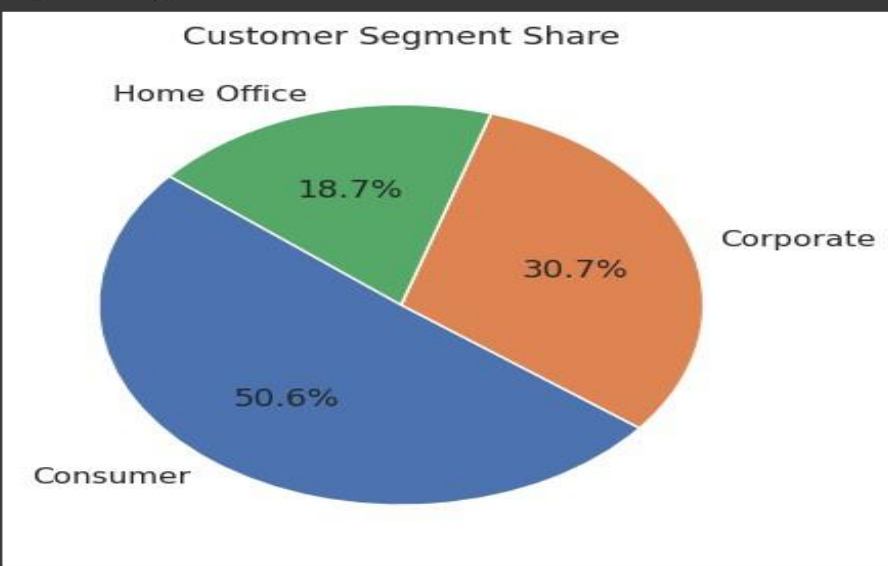
Category-wise Profit (Bar Chart)

```
sns.barplot(x='Category', y='Profit', data=df, estimator=sum)
plt.title('Category-wise Total Profit')
plt.show()
```



Segment-wise Sales (Pie Chart)

```
seg_sales = df.groupby('Segment')['Sales'].sum()
plt.pie(seg_sales, labels=seg_sales.index, autopct='%1.1f%%', startangle=140)
plt.title('Customer Segment Share')
plt.show()
```



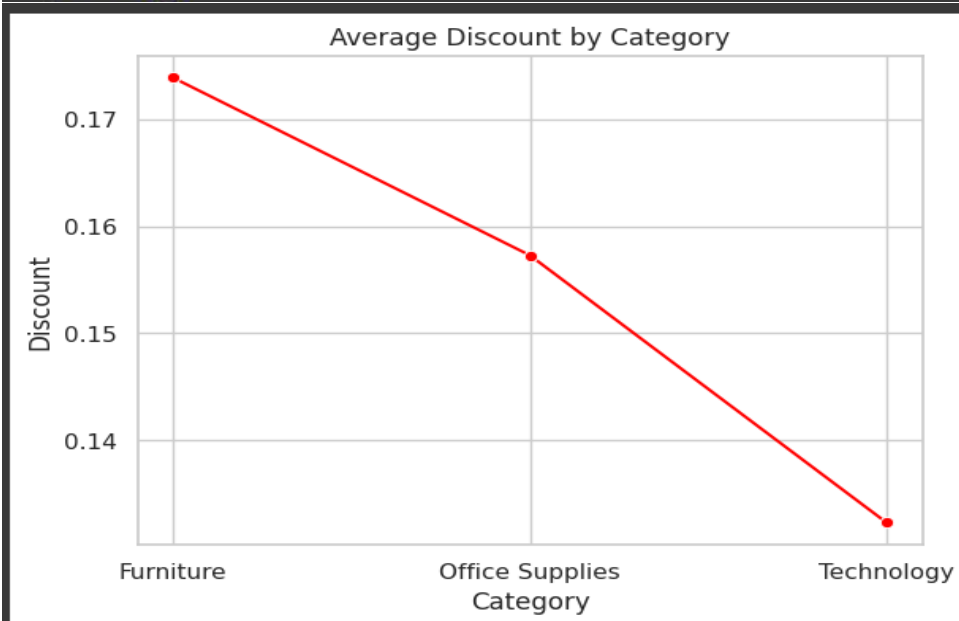
Region-wise Profit (Bar Chart)

```
region_profit = df.groupby('Region')['Profit'].sum().sort_values(ascending=False)
sns.barplot(x=region_profit.index, y=region_profit.values, palette='magma')
plt.title('Region-wise Profit')
plt.show()
```



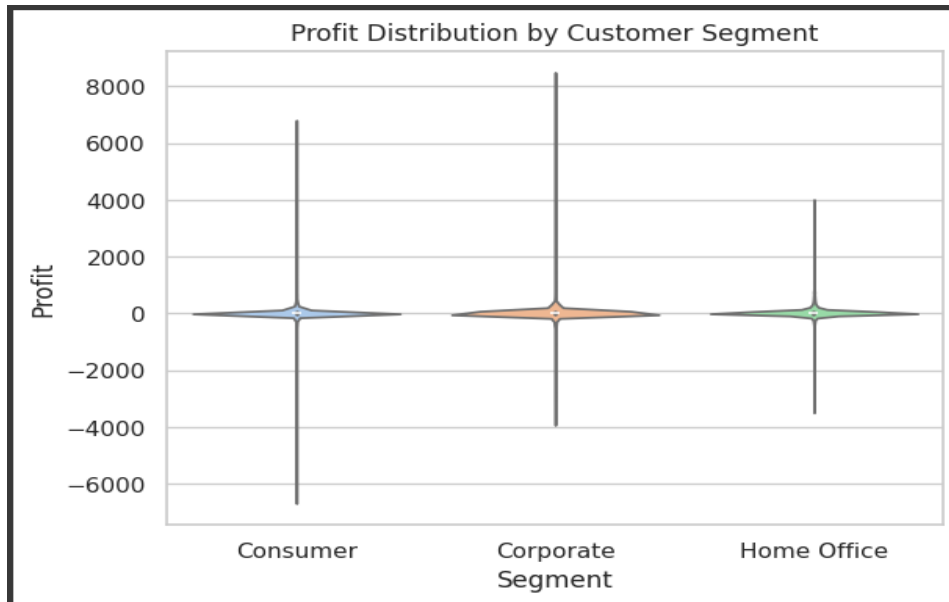
Average Discount by Category (Line Plot)

```
cat_discount = df.groupby('Category')['Discount'].mean().reset_index()
sns.lineplot(x='Category', y='Discount', data=cat_discount, marker='o', color='red')
plt.title('Average Discount by Category')
plt.show()
```



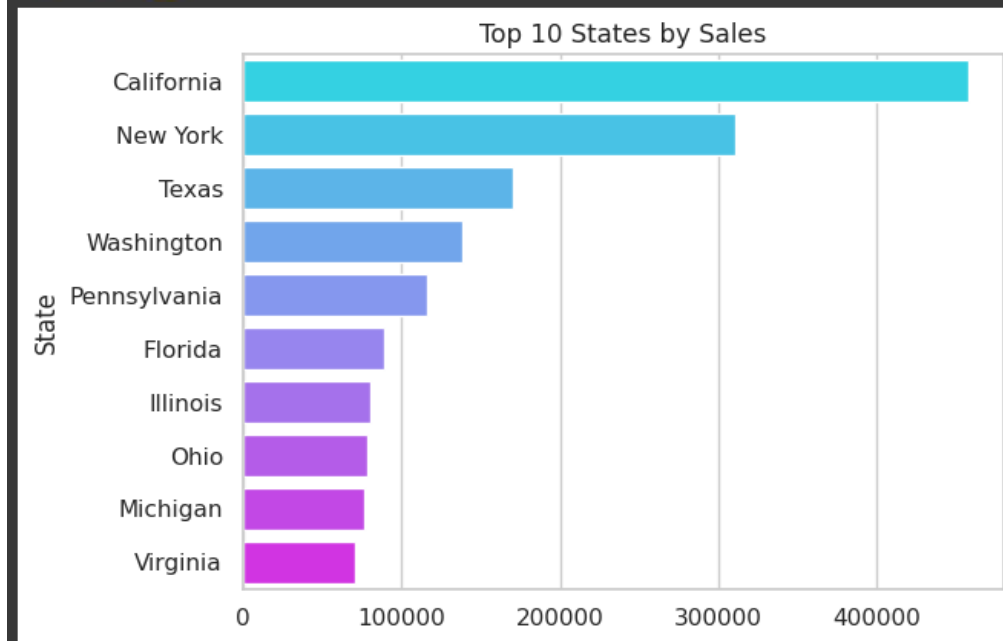
Violin Plot - Profit by Segment

```
sns.violinplot(x='Segment', y='Profit', data=df, palette='pastel')
plt.title('Profit Distribution by Customer Segment')
plt.show()
```



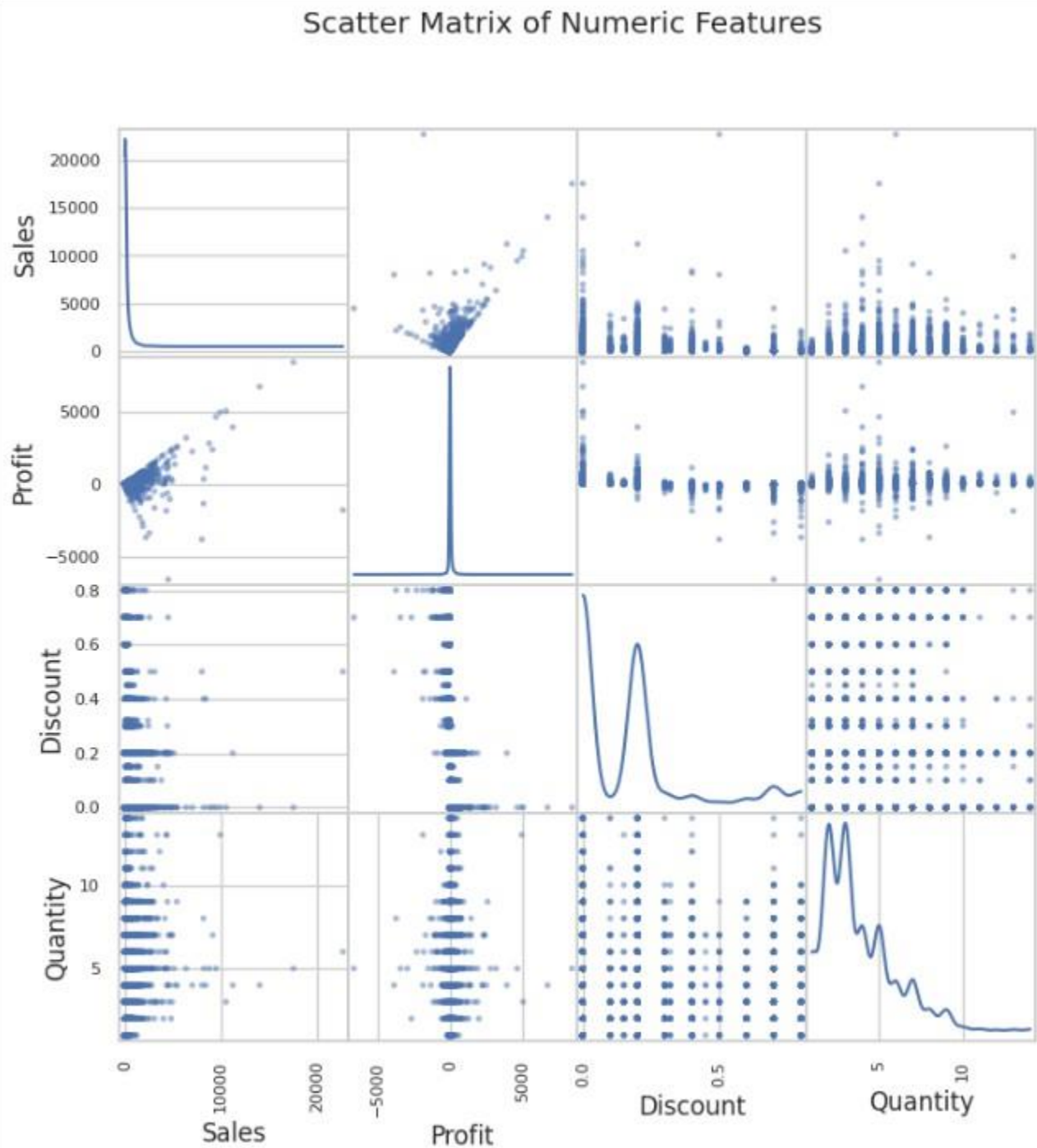
Top 10 States by Sales (Bar)

```
top_states = df.groupby('State')['Sales'].sum().sort_values(ascending=False).head(10)
sns.barplot(y=top_states.index, x=top_states.values, palette='cool')
plt.title('Top 10 States by Sales')
plt.show()
```



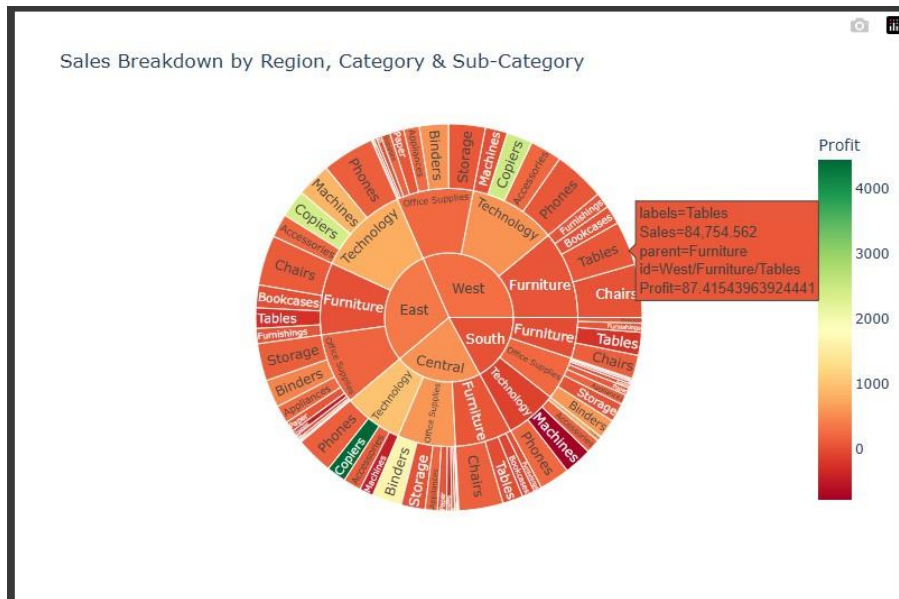
Scatter Matrix

```
scatter_matrix(df[num_cols], figsize=(8,8), diagonal='kde')  
plt.suptitle('Scatter Matrix of Numeric Features')  
plt.show()
```



Sunburst Chart - Region → Category → Sub-Category

```
fig = px.sunburst(df, path=['Region', 'Category', 'Sub-Category'], values='Sales',
                  color='Profit', color_continuous_scale='RdYlGn',
                  title='Sales Breakdown by Region, Category & Sub-Category')
fig.show()
```



Bubble Chart - Sales vs Profit vs Quantity

```
fig2 = px.scatter(df, x='Sales', y='Profit', size='Quantity', color='Category',
                  hover_name='Sub-Category', title='Sales vs Profit (Bubble = Quantity)')
fig2.show()
```



EXPORT CLEANED DATA FOR POWER BI

```
df.to_csv("Cleaned_Superstore_Data.csv", index=False)  
print("\n📁 Cleaned dataset exported for Power BI visualization.")
```

```
📁 Cleaned dataset exported for Power BI visualization.
```